# Big Data in Industry - Take-Home Assignment 1: Weekly Business Review

Olist Brazilian E-Commerce dataset (99,441 orders, Sep 2016 - Oct 2018). All numbers are produced by the SQL in the repository (`sql/`) on the SQLite database built by `run_pipeline.py`.

## 1. Pipeline and data-quality findings (Parts A + B)

The pipeline loads the 9 CSVs as **raw_*** tables (replaced on every run, so running it twice gives identical row counts - tested), builds typed/de-duplicated **stg_*** tables and one **mart_orders** table with one row per order (99,441 rows: totals, delivery dates, delay, review score). Items, payments and reviews are summed to order level *before* joining, so no order is double counted: mart revenue equals the raw item total to the cent (R$ 13,591,643.70). A missing file or column stops the run with a clear message (exit 1); a failing quality check stops it with exit 2. Each check has a *before* query (raw data) and an *after* query (cleaned data); the alarm is on the second one.

| Check | Fail in raw | Fail after cleaning | Result | How it was handled, and why |
|---|---|---|---|---|
| order id null or duplicate | 0 | 0 | PASS | DROPPED - rows with a blank order_id are removed, repeated order_ids are reduced to one row |
| delivered before purchase | 0 | 0 | PASS | FIXED - an impossible delivery timestamp is set to NULL (order kept, delivery metrics ignore it) |
| payment total mismatch | 249 | 249 | PASS | FLAGGED - kept, mart_orders.payment_mismatch = 1. Revenue uses item prices, not payments, so these rows do not distort the revenue numbers. Alarm only if the rate gets unusually high. |
| orders without customer | 0 | 0 | PASS | DROPPED - an order we cannot attach to a customer/state is useless for the analysis |
| delivered orders without items | 0 | 0 | PASS | FLAGGED - kept in mart_orders (n_items = 0, revenue NULL); they add nothing to revenue. Own check: a delivered order must contain something, so any hit raises the alarm. |
| review duplicates or bad score | 551 | 0 | PASS | DROPPED/DEDUPED - reviews with a score outside 1..5 are removed; when an order has several reviews only the most recent is kept (otherwise joins double count the order) |
| delivered status without date | 8 | 8 | PASS | FLAGGED - kept, but delivery_days / delay_days are NULL so they are skipped in delivery metrics |
| nonpositive item price | 0 | 0 | PASS | DROPPED if it ever happens (none expected) - an item with price <= 0 or negative freight would corrupt revenue |
| delivery timeline out of order | 189 | 189 | PASS | FLAGGED - kept. The order happened and was paid, only a timestamp (carrier pickup) looks wrong. The delivery metrics use purchase -> customer delivery, not the carrier timestamp. Alarm only above 0.5% of orders. |

**Reading the table.** This public dataset is already fairly clean: no null/duplicate order ids, no orders without a customer, no delivery before purchase and no non-positive prices. The real problems are (a) **551 surplus reviews** (orders with 2+ reviews) which would silently duplicate orders in a join, so we keep the latest review per order; (b) **249 orders (0.25%)** where payments differ from items + freight by more than R$ 1 (vouchers, instalment rounding) - flagged, not dropped, because revenue is measured from item prices; (c) **189 orders** whose carrier timestamp is out of order and **8 'delivered' orders without a delivery date** - flagged and ignored by delivery metrics. Tolerances (max_allowed) are written in each check file. Customer identity: `customer_id` is created per order; `customer_unique_id` identifies the person.

## 2. SQL results (Part C) - full queries in sql/analysis/q1..q5

**Q1. Month-over-month revenue and growth %** (delivered orders, item prices; LAG window). 2016 months are omitted here (tiny base, e.g. +29,777% in Oct 2016 - see the .sql output); n/m = not meaningful.

| Month | Revenue (R$) | MoM % | Month | Revenue (R$) | MoM % |
|---|---|---|---|---|---|
| 2017-01 | 111,798 | n/m | 2017-11 | 987,765 | +52.4 |
| 2017-02 | 234,223 | +109.5 | 2017-12 | 726,033 | -26.5 |
| 2017-03 | 359,199 | +53.4 | 2018-01 | 924,645 | +27.4 |
| 2017-04 | 340,670 | -5.2 | 2018-02 | 826,437 | -10.6 |
| 2017-05 | 489,338 | +43.6 | 2018-03 | 953,356 | +15.4 |
| 2017-06 | 421,923 | -13.8 | 2018-04 | 973,534 | +2.1 |
| 2017-07 | 481,605 | +14.1 | 2018-05 | 977,545 | +0.4 |
| 2017-08 | 554,700 | +15.2 | 2018-06 | 856,078 | -12.4 |
| 2017-09 | 607,400 | +9.5 | 2018-07 | 867,953 | +1.4 |
| 2017-10 | 648,248 | +6.7 | 2018-08 | 838,577 | -3.4 |

Revenue grew about 9x from Jan 2017 (R$ 112k) to the 2017-11 peak (Black Friday, R$ 988k), but in 2018 it moved sideways between R$ 0.83M and R$ 0.98M per month: growth has stalled.

**Q2. Top-3 categories per customer state** (DENSE_RANK; the 6 biggest states shown, all 27 states are in the query output).

| State | #1 category (R$) | #2 category (R$) | #3 category (R$) |
|---|---|---|---|
| SP | bed_bath_table (472,238) | health_beauty (453,916) | watches_gifts (421,547) |
| RJ | watches_gifts (174,895) | bed_bath_table (144,252) | health_beauty (140,236) |
| MG | health_beauty (154,324) | bed_bath_table (128,760) | watches_gifts (121,622) |
| PR | sports_leisure (57,653) | watches_gifts (57,287) | health_beauty (53,730) |
| RS | bed_bath_table (60,041) | furniture_decor (53,775) | computers_accessories (52,380) |
| BA | health_beauty (49,955) | watches_gifts (45,632) | sports_leisure (37,010) |

The same few categories (health_beauty, sports_leisure, watches_gifts) lead in most states, so the category mix is similar nationwide.

**Q3. 7-day rolling average of daily order count** (calendar built with a recursive CTE so days with zero orders count as 0 and the window is really 7 days).

| Statistic | Value |
|---|---|
| Highest 7-day average | 493.6 orders/day (week ending 2017-11-29, Black Friday week) |
| Typical level, Jan-mid Aug 2018 | 232 orders/day (median), range 120-314 |
| Typical level, H1 2017 | 89 orders/day (median) |

**Q4. Repeat-purchase rate** (excluding canceled/unavailable orders).

| Identifier used | Customers | Repeat customers | Repeat rate |
|---|---|---|---|
| customer_unique_id (correct) | 94,990 | 2,888 | 3.04% |
| customer_id (wrong, per-order id) | 98,207 | 0 | 0.00% |

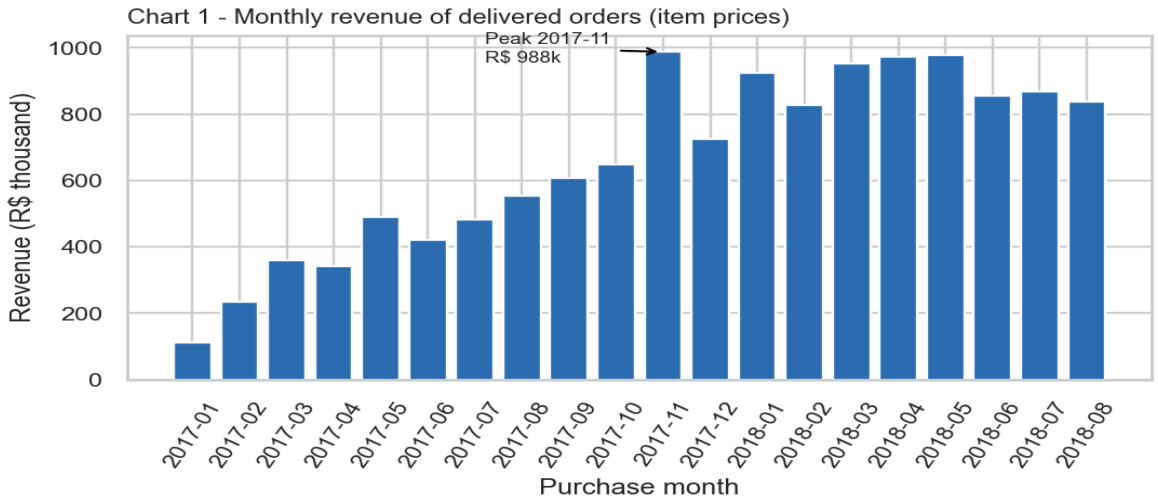
`customer_id` is regenerated for every order, so using it gives a false 0% repeat rate. Using `customer_unique_id` the true rate is **3.04%**: almost every customer buys once - retention is the weakest part of the business.

**Q5. Delivery delay versus estimate, and review score** (delay = delivered date - estimated date; negative = early). On average orders arrive **11.9 days before** the estimate, yet the few late ones are punished hard: the score falls from 4.3 (early) to 1.7 (8+ days late).

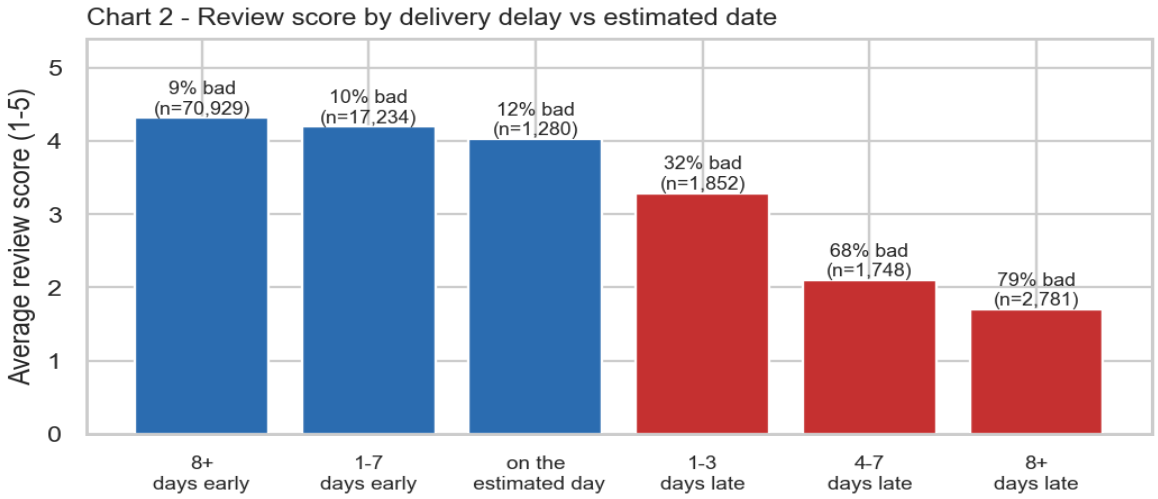
| Delivery vs estimated date | Orders | Avg delay (days) | Avg review | % 1-2 stars |
|---|---|---|---|---|
| 8+ days early | 70,929 | -15.9 | 4.32 | 9.0% |
| 1-7 days early | 17,234 | -4.8 | 4.20 | 10.2% |
| on the estimated day | 1,280 | +0.0 | 4.03 | 12.4% |
| 1-3 days late | 1,852 | +1.8 | 3.29 | 32.1% |
| 4-7 days late | 1,748 | +5.5 | 2.10 | 67.7% |
| 8+ days late | 2,781 | +19.4 | 1.70 | 79.3% |
| **All delivered orders** | **95,824** | **-11.9** | **4.16** | **12.8%** |

## 3. Metrics and charts (Part D)

| Metric | Exact formula | Why it matters | Current value |
|---|---|---|---|
| **NORTH STAR: Happy Orders per month** | COUNT(orders) WHERE order_status = 'delivered' AND review_score >= 4, grouped by purchase month | Counts only orders that were both completed *and* left the customer satisfied, so it rises only if volume and experience both improve (revenue alone can grow while customers get unhappy). | 5,109 in 2018-07; 78.9% of reviewed delivered orders are happy |
| Late-delivery rate (guardrail) | delivered orders with DATE(delivered) > DATE(estimated) / delivered orders with a delivery date | Main operational driver of bad reviews (see memo). | 6.8% |
| Bad-review rate (guardrail) | reviews with score <= 2 / all reviews of delivered orders (latest review per order) | Direct measure of dissatisfaction; early warning for churn and brand damage. | 12.8% |
| Monthly revenue (GMV) | SUM(item price) of delivered orders, by purchase month (freight excluded) | Top-line growth; tells us whether the happy-order growth is real business growth. | R$ 867,953 in 2018-07; R$ 13.22M total |
| Repeat-purchase rate | customers (customer_unique_id) with >= 2 non-canceled orders / all customers | Cheapest growth lever: a returning customer costs no acquisition spend. | 3.04% |

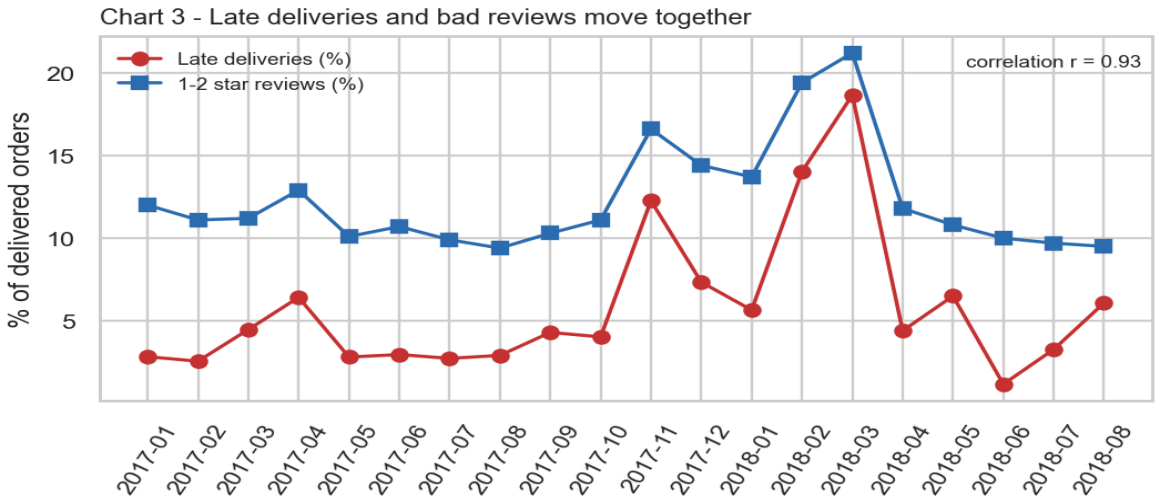



Revenue climbed through 2017 (Black Friday peak) and has plateaued since January 2018.

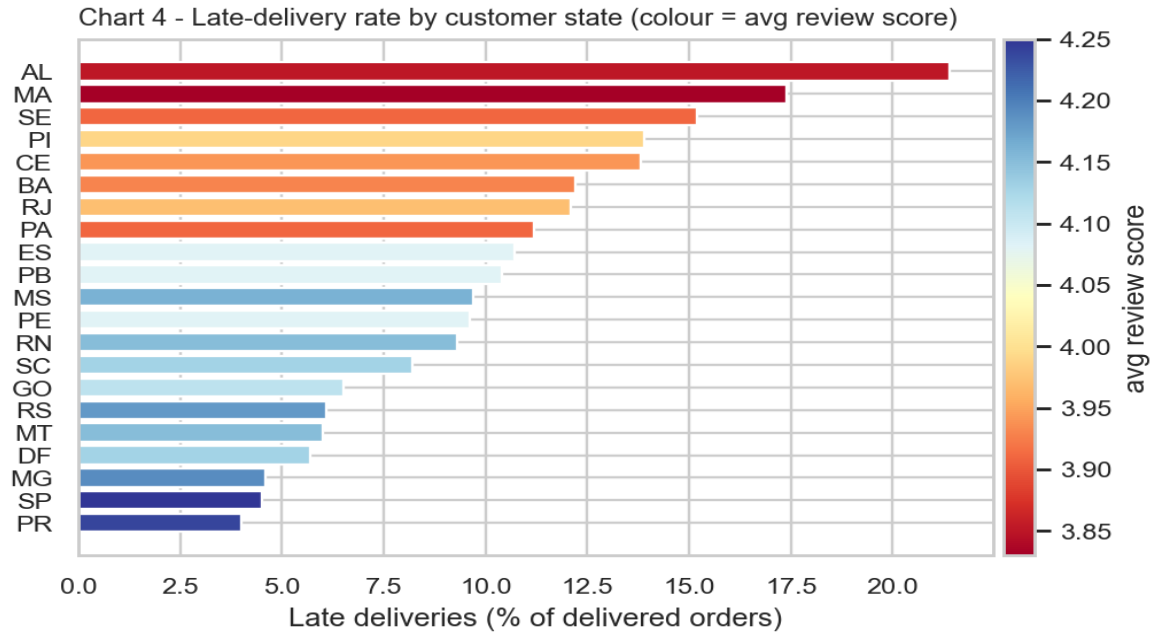



Satisfaction does not fall gradually: it collapses once an order is more than ~3 days late.

## 4. Where the problem is




Nov 2017 (Black Friday), Feb 2018 and Mar 2018 had the worst late rates and, in the same months, the highest share of bad reviews.




Late rates are 3-5x higher in the north-east (AL, MA, SE, PI, CE) than in the south-east (SP, MG, PR), and average review scores are lowest where late rates are highest.

## Memo to the Vice President

**To:** VP, Customer Experience & Operations **From:** Data Analytics **Re:** What is hurting customer satisfaction - and the one thing to fix first

**Key finding.** Late delivery is the largest single, fixable driver of bad reviews. Orders that arrive after the promised date are only **6.7%** of delivered, reviewed orders but generate **32.5%** of all 1-2 star reviews. A late order gets a bad review **62%** of the time (average 2.27 stars) against **9%** for an on-time order (4.29 stars).

**Evidence.**

• **Dose-response:** 1-3 days late = 32% bad reviews, 4-7 days late = 68%, 8+ days late = 79% (Chart 2). Being early barely helps (9% bad), so the promise matters more than the speed.

• **Over time:** in a typical month 4% of orders are late and 11% of reviews are bad (medians outside the three spikes). In Nov 2017 late orders hit 12.3% (bad reviews 16.6%), in Feb-Mar 2018 14.0% and 18.6% (bad reviews 19.4% and 21.2%). Monthly correlation r = 0.93 (Chart 3).

• **Where:** late rates are AL 21%, MA 17%, SE 15% in the north-east versus SP 4.5%, PR 4.0% in the south (Chart 4). Customers there also give the lowest scores.

• **Retention:** customers whose first order arrived late come back 2.85% of the time versus 3.79% (25% fewer), on a base repeat rate of only 3.0%.

**Caveats.** This is correlation, not proof: 68% of bad reviews come from orders that were *not* late (product quality, wrong items), so lateness is not the whole story, and the retention gap is modest.

**Recommendation.** Replace the flat delivery estimate with a **lane-based promise** (customer state x month) set at the 90th percentile of actual delivery times for that lane, and send a proactive delay message the moment an order is forecast to miss it. Orders already arrive 12 days early on average, so the current promise is too loose in fast lanes and too tight in slow lanes (north-east, Black Friday week, Feb-Mar). **Pilot:** A/B test it for 8 weeks in AL, MA, SE, PI and CE (random half of orders on the new promise). **Success** = late rate in those states below 8% (today 14-21%) and bad-review rate down at least 3 points, with checkout conversion not falling by more than 1%. **Size of the prize:** bringing the overall late rate from 6.7% to 4% would remove roughly **1,325 of the 12,272 bad reviews** (about 11%) if the late-order effect is causal - treat that as an upper bound. Owner: Logistics; first read-out after two weeks using the late-delivery and bad-review guardrails above.